EXERCISE 2

A Particular Integral by Superposition

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Superposition • Exponential Forcing

STATEMENT

> Find a particular solution for $\ddot{x} + \dot{x} - 2x = 12e^{-t} - 6e^{t}$. You may find useful parts (iii) and (iv) of the previous exercise.

PHASE 1: SETUP

The equation is $L[x] = \ddot{x} + \dot{x} - 2x = f$ with $f = 12e^{-t} - 6e^{t}$, the same operator L as items (iii) and (iv) of Exercise 1. L is linear (Robinson Eq. 11.11), so if $L[x_a] = f_a$ and $L[x_b] = f_b$ then $L[\alpha x_a + \beta x_b] = \alpha f_a + \beta f_b$ (Robinson Eq. 14.7). Hypothesis: nothing more is needed, since any single solution of $L[x] = f$ is a particular integral.

PHASE 2: FORMULATION

Rewrite the forcing in terms of the two forcings of Exercise 1:

$$f = 12e^{-t} - 6e^{t} = 4\underbrace{\left(3e^{-t}\right)}_{f_a \text{ (item iii)}} - 6\underbrace{\left(e^{t}\right)}_{f_b \text{ (item iv)}}$$

Item (iii) found x_a with $L[x_a] = 3e^{-t}$ and item (iv) found x_b with $L[x_b] = e^{t}$. Then $x_p = 4x_a - 6x_b$.

> METHOD OF THE CLASS NOTES [NOTES, P. 6]
>
> This is the decomposition rule of the educated guess in the notes: if $L[x_a] = f_a$ and $L[x_b] = f_b$, then $L[\lambda x_a + \mu x_b] = \lambda f_a + \mu f_b$ by linearity, so each piece of the forcing is guessed separately and the results are added.

PHASE 3: DERIVATION

The two particular integrals of Exercise 1

$P(k) = k^2 + k - 2 = (k-1)(k+2)$ has roots 1 and -2. For $3e^{-t}$ the exponent is $s = -1$, $P(-1) = 1 - 1 - 2 = -2 \neq 0$, so $x_a = Ke^{-t}$ and, since $\dot{x}_a = -Ke^{-t}$, $\ddot{x}_a = Ke^{-t}$: $L[x_a] = (K - K - 2K)e^{-t} = -2Ke^{-t} = 3e^{-t}$, hence $K = -\frac{3}{2}$:

$$x_a(t) = -\tfrac{3}{2}e^{-t}$$

For e^t the exponent is $s = 1$, a simple root ($P(1) = 1 + 1 - 2 = 0$), so the guess is $x_b = Kte^t$. By the product rule $\dot{x}_b = K(1+t)e^t$ and $\ddot{x}_b = K(2+t)e^t$, so

$$L[x_b] = K\left[(2+t) + (1+t) - 2t\right]e^t = 3Ke^t = e^t \implies K = \tfrac{1}{3}, \qquad x_b(t) = \tfrac{1}{3}\,t\,e^t$$

Superposition

$$x_p = 4x_a - 6x_b = 4\left(-\tfrac{3}{2}e^{-t}\right) - 6\left(\tfrac{1}{3}te^t\right) = -2te^t - 6e^{-t}$$

Direct verification by substitution

$$\dot{x}_p = -2te^t - 2e^t + 6e^{-t}$$
$$\ddot{x}_p = -2te^t - 4e^t - 6e^{-t}$$

$$\ddot{x}_p + \dot{x}_p - 2x_p = \left(-2te^t - 4e^t - 6e^{-t}\right) + \left(-2te^t - 2e^t + 6e^{-t}\right) - 2\left(-2te^t - 6e^{-t}\right) = -6e^t + 12e^{-t}$$

This is $12e^{-t} - 6e^t$, as required.

PHASE 4: INTERPRETATION AND CHECKS

RESULT

A particular solution is $x_p(t) = -2te^t - 6e^{-t}$.

The general solution is $x(t) = c_1e^t + c_2e^{-2t} - 6e^{-t} - 2te^t$ (the roots 1 and -2 of P give the complementary function).

- The term $-2te^t$ carries the extra factor t because e^t solves the homogeneous equation (resonance); the term $-6e^{-t}$ does not.

- Particular integrals are not unique: adding any multiple of e^{t} or e^{-2t} gives another one; the answer above is the one with no such terms.
- Dimensions: with $[t] = \mathrm{s}$, each term of $L[x]$ has the units of f; both sides are exponentials of a dimensionless argument $t/(1\ \mathrm{s})$, so the check is purely numerical.